Differences in Electromyographic Activity of Biceps Brachii during Barbell vs Dumbbell Curls

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Abstract & Overview

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1. Biomechanics of the Biceps Curl

The biceps curl is an open-chain isolation exercise targeting the biceps brachii (long and short heads), brachialis, and brachioradialis.
Key joint angles:
- Starting / Lowered Phase: Elbow angle is extended at 160° to 180°. Upper arms remain stationary and perpendicular to the floor.
- Peak Contraction / Up Phase: Elbow angle flexes to < 50° with full forearm supination.

2. Common Form Errors

• Error A: Torso Swinging / Momentum ("SWINGING")
  - Biomechanical Cause: The trunk tilts backward > 15° from the vertical plane during the concentric phase.
  - Root Causes: Lifting excessive weight, using lumbar hyperextension and hip thrust to bypass the sticking point.
  - Risk: Strain on the lower lumbar spine and drastically diminished biceps muscle tension.
  - Correction: Keep knees slightly soft, tuck the pelvis, brace the core, and pin shoulder blades back. The torso should remain completely static throughout the curl.
• Error B: Elbow Drifting Forward / Outward ("ELBOW DRIFTING")
  - Biomechanical Cause: Elbow joint moves forward relative to the shoulder seam (drift > 0.06 normalized units).
  - Impact: Involves the anterior deltoid to assist the lift, shortening the moment arm on the elbow and robbing the biceps of peak tension at the top.
  - Correction: Imagine elbows are glued or bolted to your ribcage. Only forearms should rotate around the stationary elbow pivot.

3. Recommended Camera Setup for Biceps Curls

- Angle: Frontal or 45-degree angle.
- Distance: 1.5 to 2.5 meters.
- Framing: Should capture from mid-thigh up through the top of the head so both upper arm alignment and torso sway can be detected.